

Initiating Search

November 4, 2025, 11:40 AM

• Search:

CAS Lexicon Search:

Hiyama coupling reaction, Hiyama-Denmark - Preferred Concept

Hiyama coupling reaction, Hiyama-Denmark - Broader Concepts (1)

Hiyama coupling reaction

Search Tasks

Task	Result Type	View
Returned Reference Results from CAS Lexicon (381)	 References	View Results
Exported: Retrieved Related Reaction Results (938)	 Reactions	View Results

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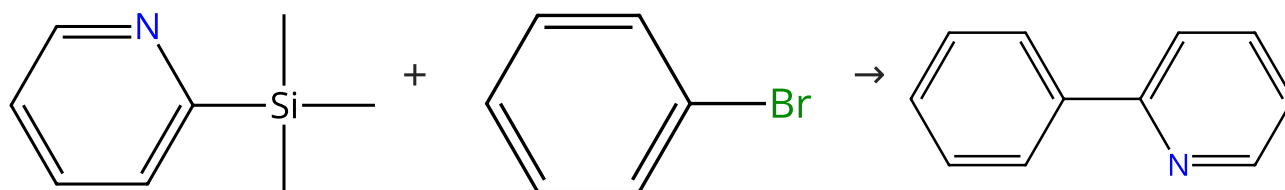
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Reactions (50)

[View in CAS SciFinder](#)

Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (60)

Suppliers (81)

Suppliers (90)

31-055-CAS-10363137

Steps: 1 Yield: 100%

Palladium(0)-catalyzed cross-coupling of 2-trimethylsilylpyridine with aryl halides

1.1 Reagents: Potassium fluoride

Catalysts: Palladium diacetate, Cuprous iodide, CataCXium A

Solvents: Dimethylformamide; 12 h, 90 °C

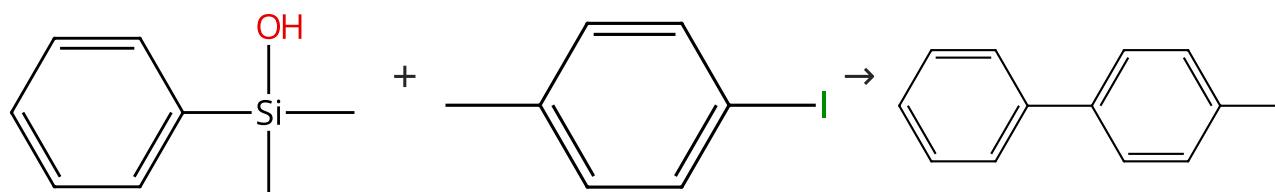
By: Blakemore, David C.; et al

Tetrahedron Letters (2011), 52(32), 4192-4195.

Experimental Protocols

Scheme 2 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (56)

Suppliers (81)

Suppliers (82)

31-055-CAS-15336808

Steps: 1 Yield: 100%

A robust method for the Hiyama-type coupling of arylsiloxanes and disiloxanes with aryl halides1.1 Reagents: Silver oxide (Ag₂O)

Catalysts: Tetrakis(triphenylphosphine)palladium, Tetrabutylammonium difluorotriphenylsilicate

Solvents: Tetrahydrofuran; 30 min, 70 °C

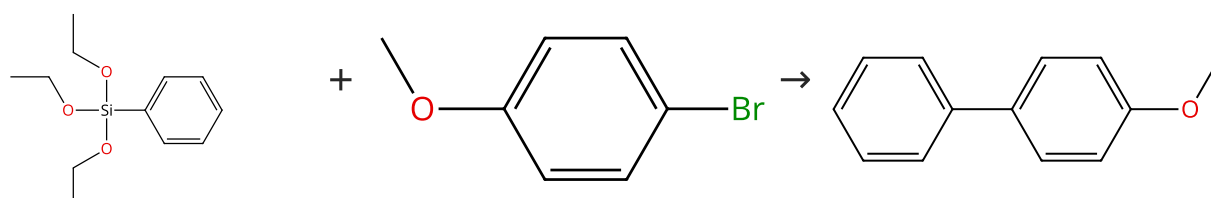
By: Napier, Spencer; et al

Tetrahedron Letters (2008), 49(24), 3939-3942.

Experimental Protocols

Scheme 3 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (75)

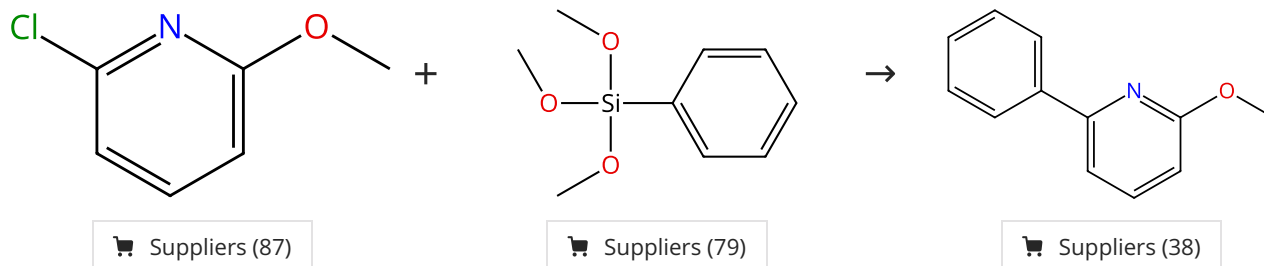
Suppliers (71)

Suppliers (69)

31-055-CAS-8301014 Steps: 1 Yield: 100% 1.1 Reagents: Tetrabutylammonium fluoride Solvents: Tetrahydrofuran; 1 h, rt 1.2 Catalysts: Bis[2-[[bis(phenylmethyl)amino-κM]methyl]phenyl-κC]di-μ-bromodipalladium Solvents: Dimethylformamide; rt $\rightarrow$ 90 °C; 3 min, 90 °C Experimental Protocols	Application of a dimeric ortho-palladated complex of tribenzylamine as an efficient catalyst in microwave-assisted Miyaura coupling reactions By: Hajipour, Abdol R.; et al Applied Organometallic Chemistry (2012), 26(2), 51-55.
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Scheme 4 (1 Reaction)

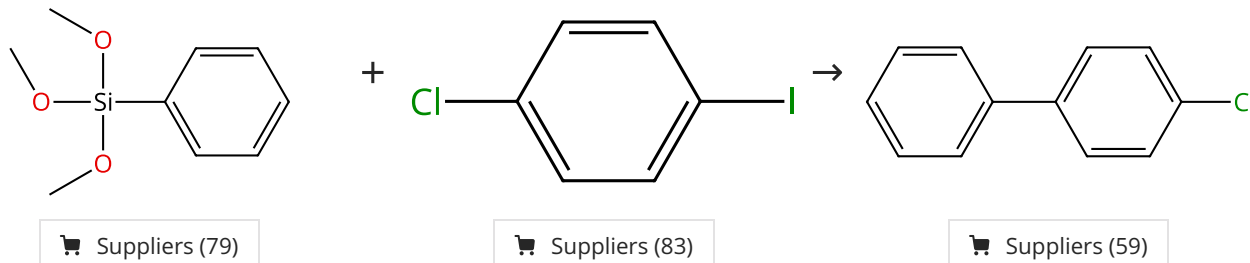
Steps: 1 Yield: 99%



31-055-CAS-21225758 Steps: 1 Yield: 99% 1.1 Catalysts: Palladium diacetate, 3-(Dicyclohexylphosphino)-2-(2-methoxyphenyl)-1-methyl-1<i>H</i>-indole Solvents: Tetrahydrofuran; 1 min, rt 1.2 Reagents: 1-Butanaminium, <i>N,N,N</i>-tributyl-, fluoride, hydrate (1:1:3); 3 h, 110 °C Experimental Protocols	A General Palladium-Catalyzed Miyaura Cross-Coupling Reaction of Aryl and Heteroaryl Chlorides By: Yuen, On Ying; et al Chemistry - A European Journal (2016), 22(19), 6471-6476.
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Scheme 5 (1 Reaction)

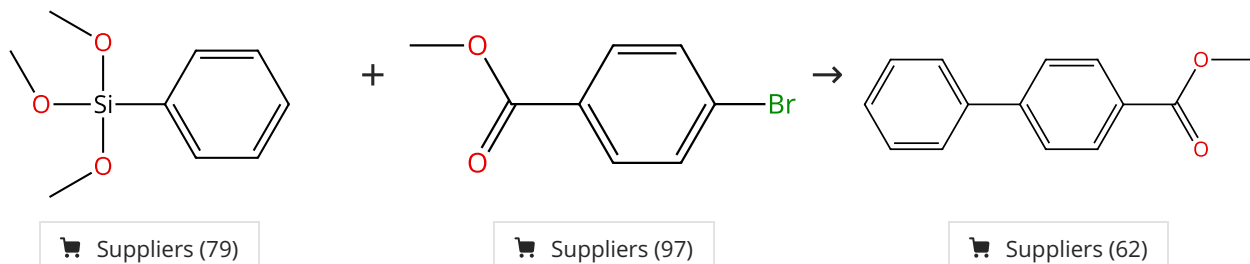
Steps: 1 Yield: 99%



31-055-CAS-1609843 Steps: 1 Yield: 99% 1.1 Reagents: Tetrabutylammonium fluoride Catalysts: Palladium (complexes with resin-supported diphenylphosphine and (tert-butyl)phenyl...) Solvents: Toluene, Water; 20 h, 100 °C Experimental Protocols	A simple and efficient reusable polystyrene-supported palladium catalyst for Miyaura cross-coupling By: Diebold, Carine; et al Tetrahedron (2013), 69(1), 264-267.
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Scheme 6 (1 Reaction)

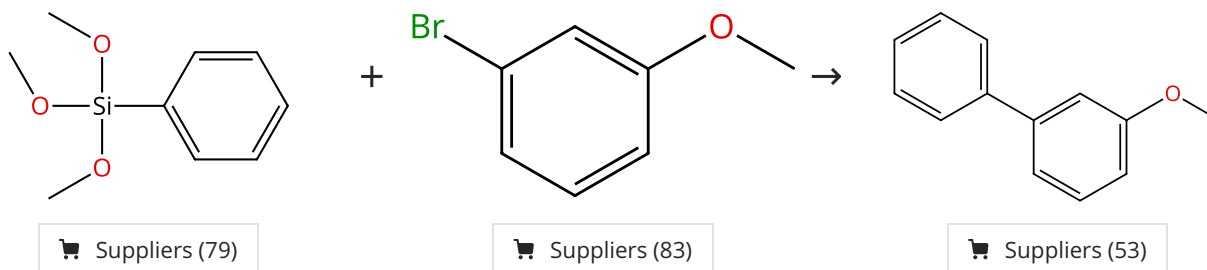
Steps: 1 Yield: 99%



31-055-CAS-13556907 Steps: 1 Yield: 99% 1.1 Reagents: Tetrabutylammonium fluoride Catalysts: Chlorodiphenylphosphine (silica bound), Palladium acetylacetonate Solvents: <i>p</i>-Xylene; 3 h, 120 °C	Silica-supported palladium-catalyzed Hiyama cross-coupling reactions using continuous flow system By: Yang, Ga Ram; et al Bulletin of the Korean Chemical Society (2010), 31(1), 250-252.
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Scheme 7 (1 Reaction)

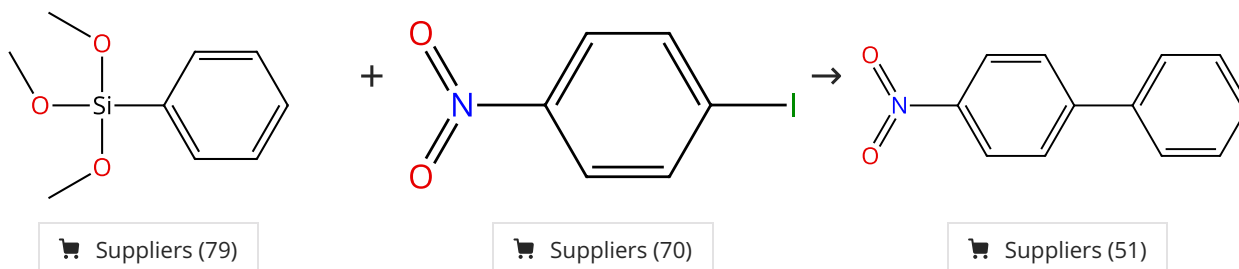
Steps: 1 Yield: 99%



31-055-CAS-18312179 Steps: 1 Yield: 99% 1.1 Reagents: Sodium hydroxide Catalysts: 1661862-34-5 Solvents: 1,4-Dioxane, Water; 4 h, 80 °C Experimental Protocols	Synthesis of adamantyl substituted 1,2,3-triazol-5-ylidene ligands and their PEPPSI-type palladium complexes By: Mitsui, Takashi; et al Tetrahedron (2015), 71(10), 1509-1514.
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Scheme 8 (1 Reaction)

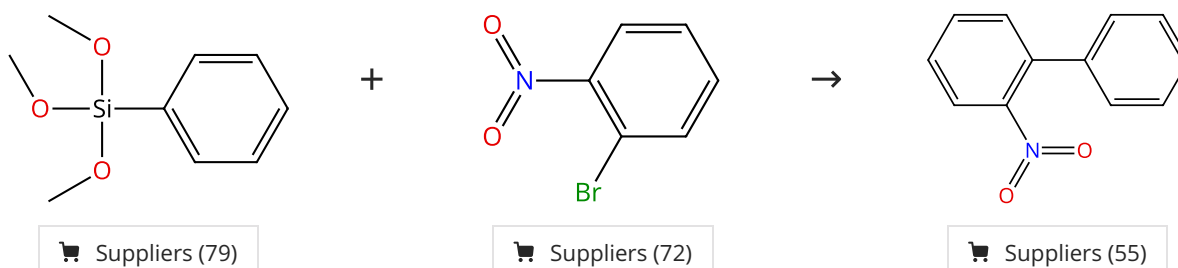
Steps: 1 Yield: 99%



31-055-CAS-1388895 Steps: 1 Yield: 99% 1.1 Reagents: Tetrabutylammonium fluoride Catalysts: Tri-<i>o</i>-tolylphosphine, Palladium, bis(acetonitrile) dichloro-; 45 min, 80 °C Experimental Protocols	Improved procedure for palladium-catalyzed Hiyama cross-coupling reaction of aryl halides with aryltri methoxysilanes under solvent-free conditions By: Li, Jin-Heng; et al Synthesis (2006), (6), 969-974.
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Scheme 9 (2 Reactions)

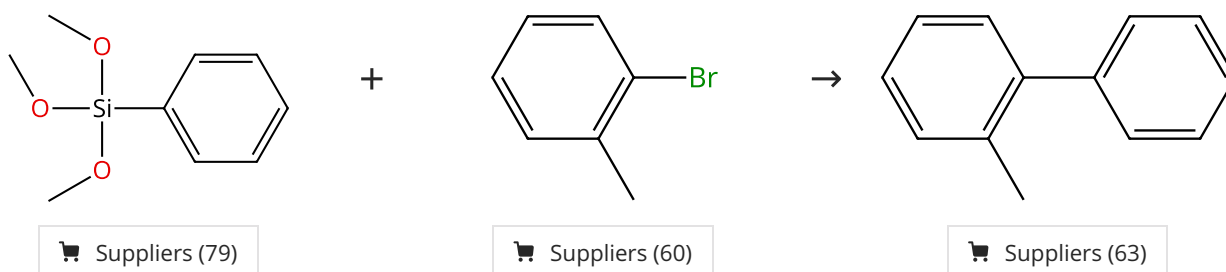
Steps: 1 Yield: 99%



31-055-CAS-14727074 Steps: 1 Yield: 99% 1.1 Reagents: Sodium hydroxide Catalysts: (SP-4-1)-[1,3-Bis(2,4,6-trimethylphenyl)-2-imidazolidinylidene]dichloro(pyridine)palladium Solvents: 1,4-Dioxane, Water; 16 h, 80 °C Experimental Protocols	Fluoride-Free Hiyama and Copper- and Amine-Free Sonogashira Coupling in Air in a Mixed Aqueous Medium by a Series of PEPPSI-Themed Precatalysts By: Dash, Chandrakanta; et al European Journal of Inorganic Chemistry (2009), (12), 1608-1618.
31-055-CAS-1328310 Steps: 1 Yield: 99% 1.1 Reagents: Sodium hydroxide Catalysts: (SP-4-1)-[1,3-Bis(2,6-dimethylphenyl)-2-imidazolidinylidene]dichloro(pyridine)palladium Solvents: 1,4-Dioxane, Water; 16 h, 80 °C Experimental Protocols	Fluoride-Free Hiyama and Copper- and Amine-Free Sonogashira Coupling in Air in a Mixed Aqueous Medium by a Series of PEPPSI-Themed Precatalysts By: Dash, Chandrakanta; et al European Journal of Inorganic Chemistry (2009), (12), 1608-1618.

Scheme 10 (1 Reaction)

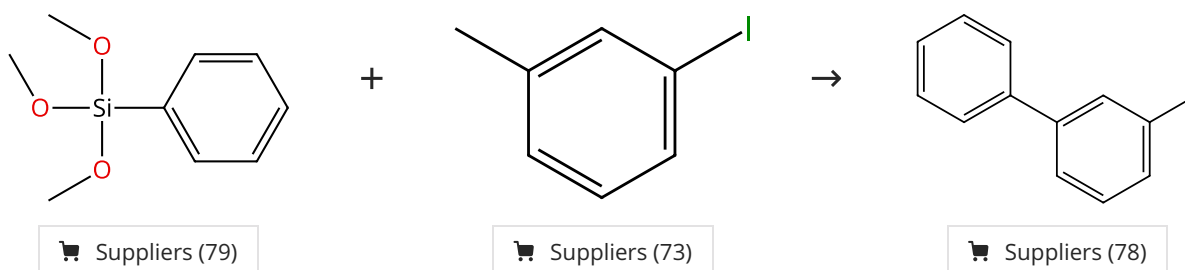
Steps: 1 Yield: 99%



31-055-CAS-18272602 Steps: 1 Yield: 99% 1.1 Reagents: Sodium hydroxide Catalysts: 1661862-34-5 Solvents: 1,4-Dioxane, Water; 4 h, 80 °C Experimental Protocols	Synthesis of adamantyl substituted 1,2,3-triazol-5-ylidene ligands and their PEPPSI-type palladium complexes By: Mitsui, Takashi; et al Tetrahedron (2015), 71(10), 1509-1514.
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Scheme 11 (1 Reaction)

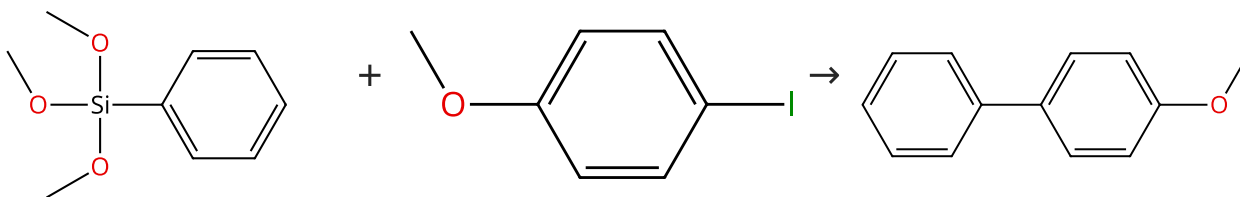
Steps: 1 Yield: 99%



31-055-CAS-6505099 Steps: 1 Yield: 99% 1.1 Reagents: Tetrabutylammonium fluoride Catalysts: Palladium (complexes with resin-supported diphenylphosphine and (tert-butyl)phenyl...) Solvents: Toluene, Water; 20 h, 100 °C Experimental Protocols	A simple and efficient reusable polystyrene-supported palladium catalyst for Hiyama cross-coupling By: Diebold, Carine; et al Tetrahedron (2013), 69(1), 264-267.
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Scheme 12 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (79)

Suppliers (88)

Suppliers (69)

31-055-CAS-7793369

Steps: 1 Yield: 99%

Improved procedure for palladium-catalyzed Hiyama cross-coupling reaction of aryl halides with aryltri methoxysilanes under solvent-free conditions

By: Li, Jin-Heng; et al

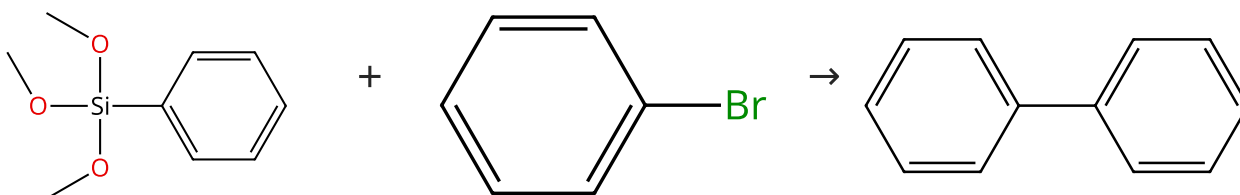
Synthesis (2006), (6), 969-974.

1.1 **Reagents:** Tetrabutylammonium fluoride
Catalysts: Tri-*o*-tolylphosphine, Palladium, bis(acetonitrile) dichloro-; 45 min, 80 °C

Experimental Protocols

Scheme 13 (4 Reactions)

Steps: 1 Yield: 98-99%



Suppliers (79)

Suppliers (81)

Suppliers (116)

31-055-CAS-13264264

Steps: 1 Yield: 99%

Silica-supported palladium-catalyzed Hiyama cross-coupling reactions using continuous flow system

By: Yang, Ga Ram; et al

Bulletin of the Korean Chemical Society (2010), 31(1), 250-252.

1.1 **Reagents:** Tetrabutylammonium fluoride
Catalysts: Chlorodiphenylphosphine (silica bound), Palladium acetylacetonate
Solvents: *p*-Xylene; 3 h, 120 °C

31-055-CAS-5320900

Steps: 1 Yield: 99%

Improved procedure for palladium-catalyzed Hiyama cross-coupling reaction of aryl halides with aryltri methoxysilanes under solvent-free conditions

By: Li, Jin-Heng; et al

Synthesis (2006), (6), 969-974.

1.1 **Reagents:** Tetrabutylammonium fluoride
Catalysts: Tri-*o*-tolylphosphine, Palladium, bis(acetonitrile) dichloro-; 50 min, 80 °C

Experimental Protocols

31-055-CAS-3386537

Steps: 1 Yield: 98%

Phosphane-Free Hiyama Cross-Coupling of Aryl and Heteroaryl Halides Catalyzed by Palladium Nanoparticles in Ionic Liquids

By: Premi, Chanchal; et al

European Journal of Organic Chemistry (2013), 2013(24), 5493-5499.

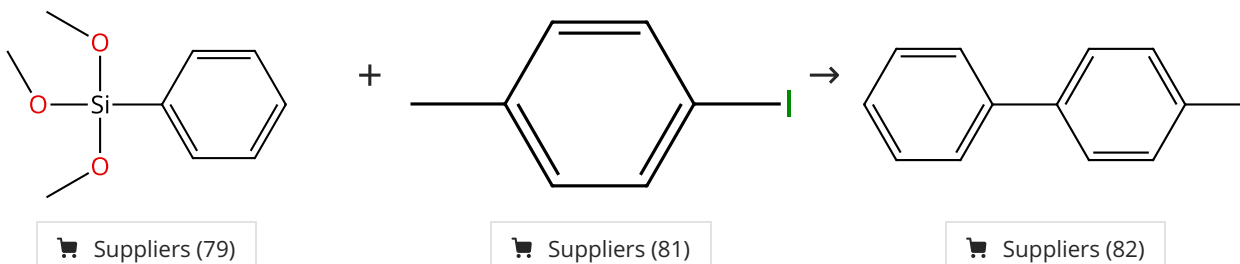
1.1 **Reagents:** 1*H*-Imidazolium, 3-butyl-1-methyl-, fluoride (1:1)
Catalysts: Palladium diacetate, 1*H*-Imidazolium, 3-(3-cyanopropyl)-1-methyl-, hexafluorophosphate(1-) (1:1)
Solvents: Acetonitrile; 8 h, 80 °C

Experimental Protocols

31-055-CAS-7899130 Steps: 1 Yield: 98% 1.1 Reagents: Potassium tetrachloropalladate, Polyethylene glycol, Ethanaminium, <i>N,N,N</i>-triethyl-, (<i>OC</i>-6-21)-acetylpentacarbonylungstate(1-) (1:1) Solvents: Water; 5 min, rt 1.2 1 min, 90 °C 1.3 Reagents: Sodium hydroxide Solvents: Water; 2 h, 90 °C Experimental Protocols	Convenient Synthesis of Palladium Nanoparticles and Catalysis of Hiyama Coupling Reaction in Water By: Srimani, Dipankar; et al Organic Letters (2007), 9(18), 3639-3642.
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Scheme 14 (1 Reaction)

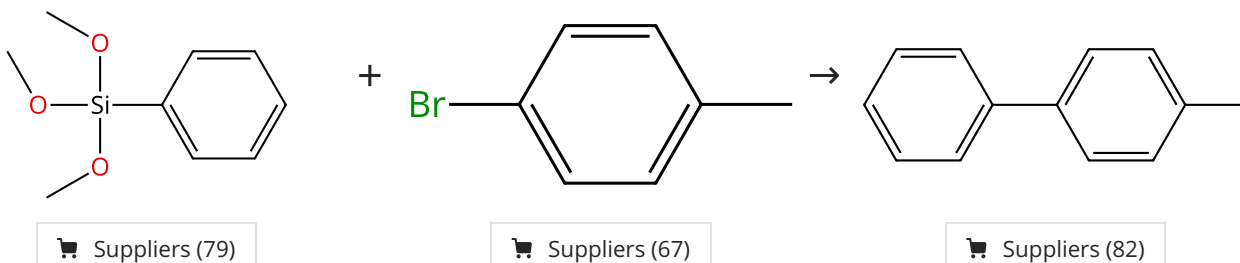
Steps: 1 Yield: 99%



31-055-CAS-4941759 Steps: 1 Yield: 99% 1.1 Reagents: Tetrabutylammonium fluoride Catalysts: Palladium (complexes with resin-supported diphenylphosphine and (tert-butyl)pheny...) Solvents: Toluene, Water; 20 h, 100 °C Experimental Protocols	A simple and efficient reusable polystyrene-supported palladium catalyst for Hiyama cross-coupling By: Diebold, Carine; et al Tetrahedron (2013), 69(1), 264-267.
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Scheme 15 (6 Reactions)

Steps: 1 Yield: 99%

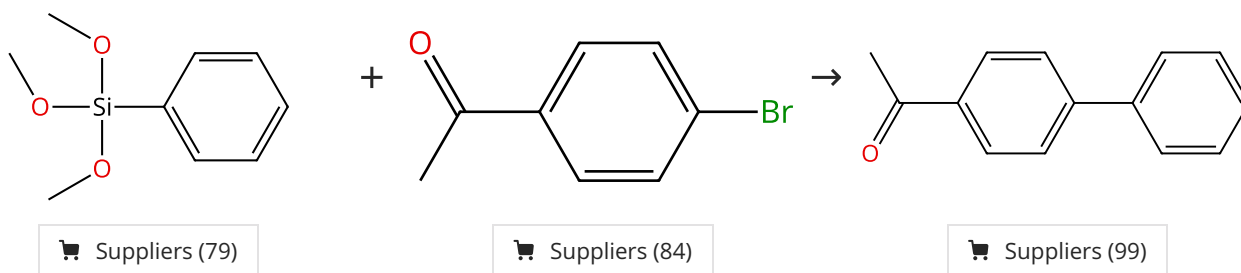


31-055-CAS-6932357 Steps: 1 Yield: 99% 1.1 Reagents: Tetrabutylammonium fluoride Catalysts: Chlorodiphenylphosphine (silica bound), Palladium acetylacetonate Solvents: <i>p</i>-Xylene; 3 h, 120 °C	Silica-supported palladium-catalyzed Hiyama cross-coupling reactions using continuous flow system By: Yang, Ga Ram; et al Bulletin of the Korean Chemical Society (2010), 31(1), 250-252.
31-055-CAS-3793378 Steps: 1 Yield: 99% 1.1 Reagents: Sodium hydroxide Catalysts: (<i>SP</i>-4-1)-Dichloro[2-[2-[(1,1-dimethylethyl)amino]-2-oxoethyl]-2,4-dihydro-4-(phenylmethyl)-3<i>H</i>-1,2,4-triazol-3-ylidene](pyridine)palladium Solvents: 1,4-Dioxane, Water; 4 h, 80 °C Experimental Protocols	Fluoride-Free Hiyama and Copper- and Amine-Free Sonogashira Coupling in Air in a Mixed Aqueous Medium by a Series of PEPPSI-Themed Precatalysts By: Dash, Chandrakanta; et al European Journal of Inorganic Chemistry (2009), (12), 1608-1618.

31-055-CAS-12924982 Steps: 1 Yield: 99% 1.1 Reagents: Sodium hydroxide Catalysts: (SP-4-1)-[1,3-Bis(2,4,6-trimethylphenyl)-2-imidazolidinylidene]dichloro(pyridine)palladium Solvents: 1,4-Dioxane, Water; 4 h, 80 °C Experimental Protocols	Fluoride-Free Hiyama and Copper- and Amine-Free Sonogashira Coupling in Air in a Mixed Aqueous Medium by a Series of PEPPSI-Themed Precatalysts By: Dash, Chandrakanta; et al European Journal of Inorganic Chemistry (2009), (12), 1608-1618.
31-055-CAS-8666579 Steps: 1 Yield: 99% 1.1 Reagents: Sodium hydroxide Catalysts: (SP-4-1)-[1,3-Bis[2,6-bis(1-methylethyl)phenyl]-2-imidazolidinylidene]dichloro(pyridine)palladium Solvents: 1,4-Dioxane, Water; 4 h, 80 °C Experimental Protocols	Fluoride-Free Hiyama and Copper- and Amine-Free Sonogashira Coupling in Air in a Mixed Aqueous Medium by a Series of PEPPSI-Themed Precatalysts By: Dash, Chandrakanta; et al European Journal of Inorganic Chemistry (2009), (12), 1608-1618.
31-055-CAS-15063244 Steps: 1 Yield: 99% 1.1 Reagents: Sodium hydroxide Catalysts: (SP-4-1)-[1,3-Bis(2,6-dimethylphenyl)-2-imidazolidinylidene]dichloro(pyridine)palladium Solvents: 1,4-Dioxane, Water; 4 h, 80 °C Experimental Protocols	Fluoride-Free Hiyama and Copper- and Amine-Free Sonogashira Coupling in Air in a Mixed Aqueous Medium by a Series of PEPPSI-Themed Precatalysts By: Dash, Chandrakanta; et al European Journal of Inorganic Chemistry (2009), (12), 1608-1618.
31-055-CAS-1675392 Steps: 1 Yield: 99% 1.1 Reagents: Sodium hydroxide Catalysts: (SP-4-1)-Dichloro[2,4-dihydro-2-(1-methylethyl)-4-(phenylmethyl)-3H-1,2,4-triazol-3-ylidene](pyridine)palladium Solvents: 1,4-Dioxane, Water; 4 h, 80 °C Experimental Protocols	Fluoride-Free Hiyama and Copper- and Amine-Free Sonogashira Coupling in Air in a Mixed Aqueous Medium by a Series of PEPPSI-Themed Precatalysts By: Dash, Chandrakanta; et al European Journal of Inorganic Chemistry (2009), (12), 1608-1618.

Scheme 16 (7 Reactions)

Steps: 1 Yield: 99%

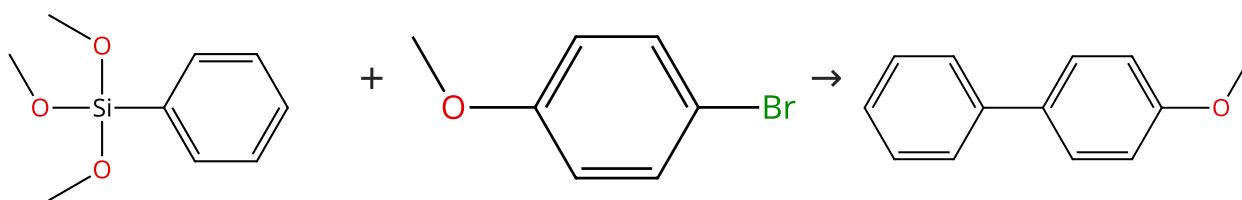


31-055-CAS-18285246 Steps: 1 Yield: 99% 1.1 Reagents: Sodium hydroxide Catalysts: 1661862-34-5 Solvents: 1,4-Dioxane, Water; 4 h, 80 °C Experimental Protocols	Synthesis of adamantyl substituted 1,2,3-triazol-5-ylidene ligands and their PEPPSI-type palladium complexes By: Mitsui, Takashi; et al Tetrahedron (2015), 71(10), 1509-1514.
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31-055-CAS-6002475 Steps: 1 Yield: 99% 1.1 Reagents: Sodium hydroxide Catalysts: Palladium, bromo[(4-carboxylato-2,6-pyridinediyl-κM)bis[3-(4-ethoxy-4-oxobutyl)-1<i>H</i>-imidazol-1-yl-2(3<i>H</i>)-ylidene-κC²]], hydrobromide (1:1), (<i>SP</i>-4-2)- Solvents: Water; 5 min, rt 1.2 Reagents: Sodium hydroxide Solvents: Water; 5 min, rt; 2 h, 160 °C	On-water Hiyama coupling catalyzed by C NC pincer complexes of Pd(II) By: Ines, Blanca; et al ARKIVOC (Gainesville, FL, United States) (2011), (3), 191-199.
31-055-CAS-12420950 Steps: 1 Yield: 99% 1.1 Reagents: Sodium hydroxide Catalysts: (<i>SP</i>-4-1)-Dichloro[2,4-dihydro-2-(1-methylethyl)-4-(phenylmethyl)-3<i>H</i>-1,2,4-triazol-3-ylidene](pyridine)palladium Solvents: 1,4-Dioxane, Water; 4 h, 80 °C Experimental Protocols	Fluoride-Free Hiyama and Copper- and Amine-Free Sonogashira Coupling in Air in a Mixed Aqueous Medium by a Series of PEPPSI-Themed Precatalysts By: Dash, Chandrakanta; et al European Journal of Inorganic Chemistry (2009), (12), 1608-1618.
31-055-CAS-3889496 Steps: 1 Yield: 99% 1.1 Reagents: Sodium hydroxide Catalysts: (<i>SP</i>-4-1)-[1,3-Bis[2,6-bis(1-methylethyl)phenyl]-2-imidazolidinylidene]dichloro(pyridine)palladium Solvents: 1,4-Dioxane, Water; 4 h, 80 °C Experimental Protocols	Fluoride-Free Hiyama and Copper- and Amine-Free Sonogashira Coupling in Air in a Mixed Aqueous Medium by a Series of PEPPSI-Themed Precatalysts By: Dash, Chandrakanta; et al European Journal of Inorganic Chemistry (2009), (12), 1608-1618.
31-055-CAS-8169408 Steps: 1 Yield: 99% 1.1 Reagents: Sodium hydroxide Catalysts: (<i>SP</i>-4-1)-[1,3-Bis(2,4,6-trimethylphenyl)-2-imidazolidinylidene]dichloro(pyridine)palladium Solvents: 1,4-Dioxane, Water; 4 h, 80 °C Experimental Protocols	Fluoride-Free Hiyama and Copper- and Amine-Free Sonogashira Coupling in Air in a Mixed Aqueous Medium by a Series of PEPPSI-Themed Precatalysts By: Dash, Chandrakanta; et al European Journal of Inorganic Chemistry (2009), (12), 1608-1618.
31-055-CAS-10300504 Steps: 1 Yield: 99% 1.1 Reagents: Sodium hydroxide Catalysts: (<i>SP</i>-4-1)-[1,3-Bis(2,6-dimethylphenyl)-2-imidazolidinylidene]dichloro(pyridine)palladium Solvents: 1,4-Dioxane, Water; 4 h, 80 °C Experimental Protocols	Fluoride-Free Hiyama and Copper- and Amine-Free Sonogashira Coupling in Air in a Mixed Aqueous Medium by a Series of PEPPSI-Themed Precatalysts By: Dash, Chandrakanta; et al European Journal of Inorganic Chemistry (2009), (12), 1608-1618.
31-055-CAS-6033097 Steps: 1 Yield: 99% 1.1 Reagents: Sodium hydroxide Catalysts: (<i>SP</i>-4-1)-[1,3-Bis(2,6-diethylphenyl)-2-imidazolidinylidene]dichloro(pyridine)palladium Solvents: 1,4-Dioxane, Water; 4 h, 80 °C Experimental Protocols	Fluoride-Free Hiyama and Copper- and Amine-Free Sonogashira Coupling in Air in a Mixed Aqueous Medium by a Series of PEPPSI-Themed Precatalysts By: Dash, Chandrakanta; et al European Journal of Inorganic Chemistry (2009), (12), 1608-1618.

Scheme 17 (9 Reactions)

Steps: 1 Yield: 98-99%



Suppliers (79)

Suppliers (71)

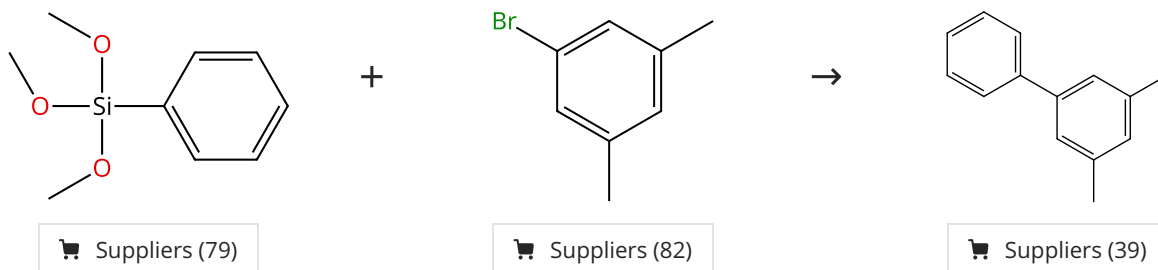
Suppliers (69)

31-055-CAS-18183543 Steps: 1 Yield: 99% 1.1 Reagents: Sodium hydroxide Catalysts: 1661862-34-5 Solvents: 1,4-Dioxane, Water; 4 h, 80 °C Experimental Protocols	Synthesis of adamantyl substituted 1,2,3-triazol-5-ylidene ligands and their PEPPSI-type palladium complexes By: Mitsui, Takashi; et al Tetrahedron (2015), 71(10), 1509-1514.
31-055-CAS-4198201 Steps: 1 Yield: 99% 1.1 Reagents: Sodium hydroxide Catalysts: Palladium diacetate, 1<i>H</i>-Imidazolium, 3-(2-hydroxy-2-phenylethyl)-1-(phenylmethyl)-, chloride (1:1) Solvents: Water; rt; 60 min, 120 °C Experimental Protocols	(NHC)Palladium complexes from hydroxy-functionalized imidazolium salts as catalyst for the microwave-accelerated fluorine-free Hiyama reaction By: Penafiel, Itziar; et al European Journal of Organic Chemistry (2011), 2011(35), 7174-7181.
31-055-CAS-6537187 Steps: 1 Yield: 99% 1.1 Reagents: Sodium hydroxide Catalysts: (<i>SP</i>-4-1)-Dichloro[2-[2-[(1,1-dimethylethyl)amino]-2-oxoethyl]-2,4-dihydro-4-(phenylmethyl)-3<i>H</i>-1,2,4-triazol-3-ylidene](pyridine)palladium Solvents: 1,4-Dioxane, Water; 4 h, 80 °C Experimental Protocols	Fluoride-Free Hiyama and Copper- and Amine-Free Sonogashira Coupling in Air in a Mixed Aqueous Medium by a Series of PEPPSI-Themed Precatalysts By: Dash, Chandrakanta; et al European Journal of Inorganic Chemistry (2009), (12), 1608-1618.
31-055-CAS-4407537 Steps: 1 Yield: 99% 1.1 Reagents: Sodium hydroxide Catalysts: (<i>SP</i>-4-1)-Dichloro[2,4-dihydro-2-(1-methylethyl)-4-(phenylmethyl)-3<i>H</i>-1,2,4-triazol-3-ylidene](pyridine)palladium Solvents: 1,4-Dioxane, Water; 4 h, 80 °C Experimental Protocols	Fluoride-Free Hiyama and Copper- and Amine-Free Sonogashira Coupling in Air in a Mixed Aqueous Medium by a Series of PEPPSI-Themed Precatalysts By: Dash, Chandrakanta; et al European Journal of Inorganic Chemistry (2009), (12), 1608-1618.
31-055-CAS-6266520 Steps: 1 Yield: 99% 1.1 Reagents: Sodium hydroxide Catalysts: (<i>SP</i>-4-1)-[1,3-Bis(2,6-diethylphenyl)-2-imidazolidinylidene]dichloro(pyridine)palladium Solvents: 1,4-Dioxane, Water; 4 h, 80 °C Experimental Protocols	Fluoride-Free Hiyama and Copper- and Amine-Free Sonogashira Coupling in Air in a Mixed Aqueous Medium by a Series of PEPPSI-Themed Precatalysts By: Dash, Chandrakanta; et al European Journal of Inorganic Chemistry (2009), (12), 1608-1618.
31-055-CAS-10523285 Steps: 1 Yield: 99% 1.1 Reagents: Sodium hydroxide Catalysts: (<i>SP</i>-4-1)-[1,3-Bis(2,6-dimethylphenyl)-2-imidazolidinylidene]dichloro(pyridine)palladium Solvents: 1,4-Dioxane, Water; 4 h, 80 °C Experimental Protocols	Fluoride-Free Hiyama and Copper- and Amine-Free Sonogashira Coupling in Air in a Mixed Aqueous Medium by a Series of PEPPSI-Themed Precatalysts By: Dash, Chandrakanta; et al European Journal of Inorganic Chemistry (2009), (12), 1608-1618.

31-055-CAS-4086579 Steps: 1 Yield: 99% 1.1 Reagents: Sodium hydroxide Catalysts: (SP-4-1)-[1,3-Bis[2,6-bis(1-methylethyl)phenyl]-2-imidazolidinylidene]dichloro(pyridine)palladium Solvents: 1,4-Dioxane, Water; 4 h, 80 °C Experimental Protocols	Fluoride-Free Hiyama and Copper- and Amine-Free Sonogashira Coupling in Air in a Mixed Aqueous Medium by a Series of PEPPSI-Themed Precatalysts By: Dash, Chandrakanta; et al European Journal of Inorganic Chemistry (2009), (12), 1608-1618.
31-055-CAS-8388194 Steps: 1 Yield: 99% 1.1 Reagents: Sodium hydroxide Catalysts: (SP-4-1)-[1,3-Bis(2,4,6-trimethylphenyl)-2-imidazolidinylidene]dichloro(pyridine)palladium Solvents: 1,4-Dioxane, Water; 4 h, 80 °C Experimental Protocols	Fluoride-Free Hiyama and Copper- and Amine-Free Sonogashira Coupling in Air in a Mixed Aqueous Medium by a Series of PEPPSI-Themed Precatalysts By: Dash, Chandrakanta; et al European Journal of Inorganic Chemistry (2009), (12), 1608-1618.
31-055-CAS-12152016 Steps: 1 Yield: 98% 1.1 Reagents: Potassium tetrachloropalladate, Polyethylene glycol, Ethanaminium, <i>N,N,N</i>-triethyl-, (<i>OC</i>-6-21)-acetylpentacarbonylungstate(1-) (1:1) Solvents: Water; 5 min, rt 1.2 1 min, 90 °C 1.3 Reagents: Sodium hydroxide Solvents: Water; 3 h, 90 °C Experimental Protocols	Convenient Synthesis of Palladium Nanoparticles and Catalysis of Hiyama Coupling Reaction in Water By: Srimani, Dipankar; et al Organic Letters (2007), 9(18), 3639-3642.

Scheme 18 (1 Reaction)

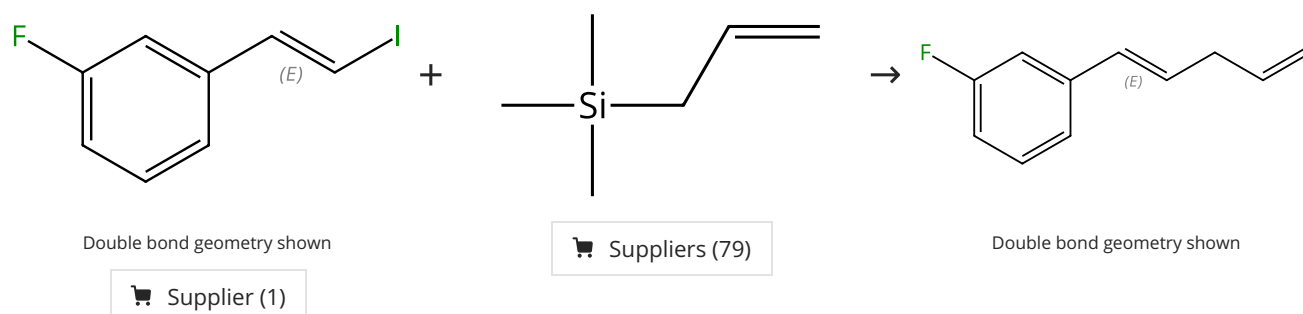
Steps: 1 Yield: 98%



31-055-CAS-9593823 Steps: 1 Yield: 98% 1.1 Reagents: Tetrabutylammonium fluoride Catalysts: Tri-<i>o</i>-tolylphosphine, Palladium, bis(acetonitrile) dichloro-; 35 min, 80 °C Experimental Protocols	Improved procedure for palladium-catalyzed Hiyama cross-coupling reaction of aryl halides with aryltri methoxysilanes under solvent-free conditions By: Li, Jin-Heng; et al Synthesis (2006), (6), 969-974.
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Scheme 19 (1 Reaction)

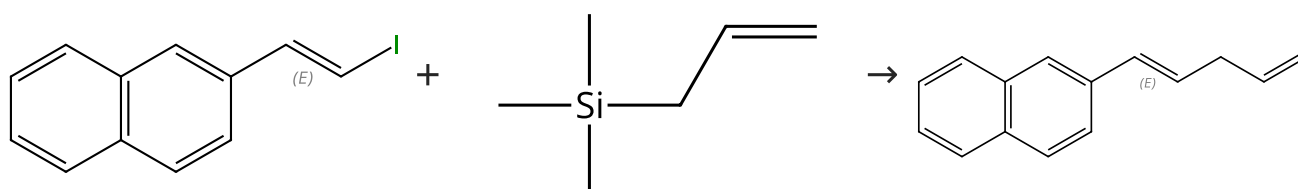
Steps: 1 Yield: 98%



31-614-CAS-42963146	Steps: 1 Yield: 98%	Hemilabile P,N-ligand-assisted gold catalyzed Hiyama allylation to access allylarenes and 1,4-diene derivatives By: He, Yuanhao; et al Organic Chemistry Frontiers (2025), 12(2), 569-576.
1.1 Reagents: Sodium bicarbonate Catalysts: Silver triflate, [2-[Bis(tricyclo[3.3.1.1 ^{3,7}]dec-1-yl)phosphino-κP]-N,N-dimethylbenzenamine]chlorogold Solvents: 1,2-Dichloroethane; 8 h, rt		
1.2 Reagents: Hydrochloric acid Solvents: Ethyl acetate; rt		
Experimental Protocols		

Scheme 20 (1 Reaction)

Steps: 1 Yield: 98%



Double bond geometry shown

Suppliers (79)

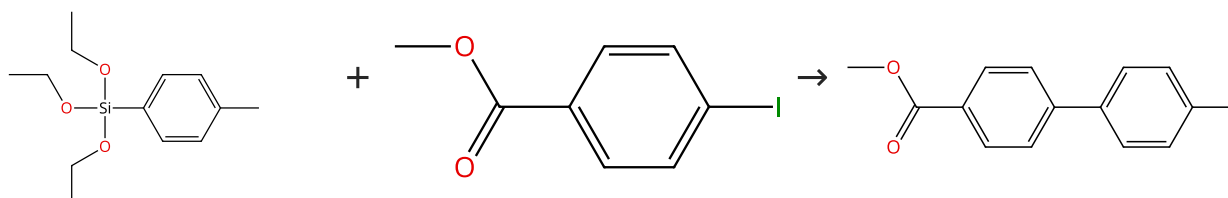
Double bond geometry shown

Suppliers (3)

31-614-CAS-42963155	Steps: 1 Yield: 98%	Hemilabile P,N-ligand-assisted gold catalyzed Hiyama allylation to access allylarenes and 1,4-diene derivatives By: He, Yuanhao; et al Organic Chemistry Frontiers (2025), 12(2), 569-576.
1.1 Reagents: Sodium bicarbonate Catalysts: Silver triflate, [2-[Bis(tricyclo[3.3.1.1 ^{3,7}]dec-1-yl)phosphino-κP]-N,N-dimethylbenzenamine]chlorogold Solvents: 1,2-Dichloroethane; 8 h, rt		
1.2 Reagents: Hydrochloric acid Solvents: Ethyl acetate; rt		
Experimental Protocols		

Scheme 21 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (48)

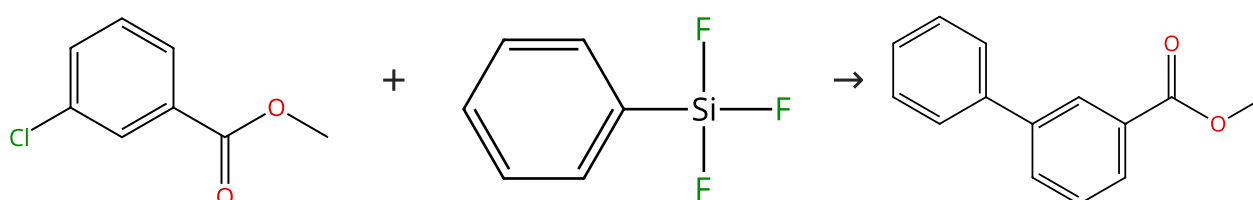
Suppliers (91)

Suppliers (53)

31-055-CAS-18106896	Steps: 1 Yield: 98%	Very efficient and broad-in-scope palladium-catalyzed Hiyama cross-coupling. The role of water and copper (I) salts By: Traficante, Carla I.; et al RSC Advances (2015), 5(34), 26796-26800.
1.1 Reagents: Tetrabutylammonium fluoride, Water Catalysts: Cuprous iodide, Tetrakis(triphenylphosphine)palladium Solvents: Tetrahydrofuran; 18 h, 80 °C		

Scheme 22 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (61)

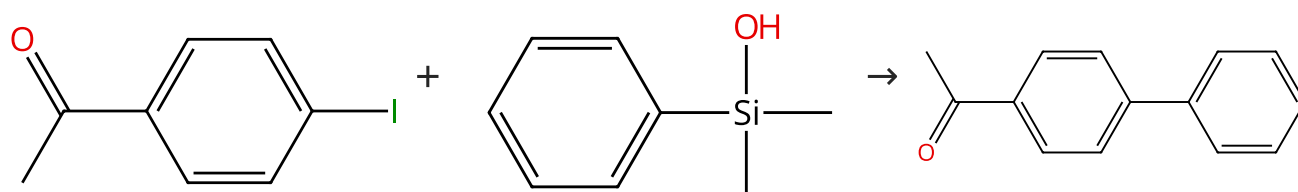
Suppliers (37)

Suppliers (29)

31-055-CAS-13307208	Steps: 1 Yield: 98%	Palladium-catalyzed Hiyama cross-coupling of aryltrifluorosilanes with aryl and heteroaryl chlorides
1.1 Reagents: Tetrabutylammonium fluoride Catalysts: Palladium diacetate, X-Phos Solvents: <i>tert</i> -Butanol; rt; 5 min, rt; rt; overnight, 60 °C		By: Molander, Gary A.; et al Journal of Organic Chemistry (2011), 76(21), 9182-9187.
Experimental Protocols		

Scheme 23 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (87)

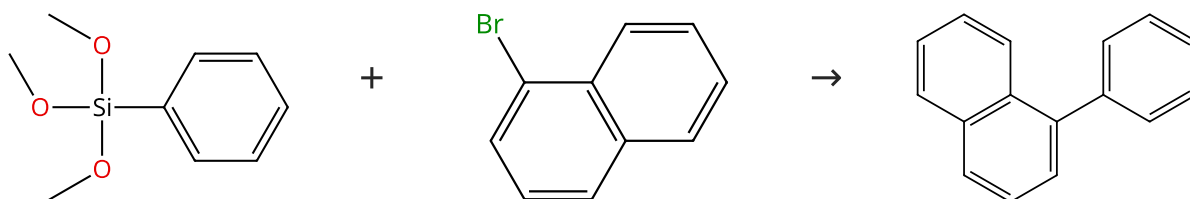
Suppliers (56)

Suppliers (99)

31-055-CAS-14613658	Steps: 1 Yield: 98%	A robust method for the Hiyama-type coupling of arylsiloxanes and disiloxanes with aryl halides
1.1 Reagents: Silver oxide (Ag ₂ O) Catalysts: Tetrakis(triphenylphosphine)palladium, Tetrabutylammonium difluorotriphenylsilicate Solvents: Tetrahydrofuran; 30 min, 70 °C		By: Napier, Spencer; et al Tetrahedron Letters (2008), 49(24), 3939-3942.
Experimental Protocols		

Scheme 24 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (79)

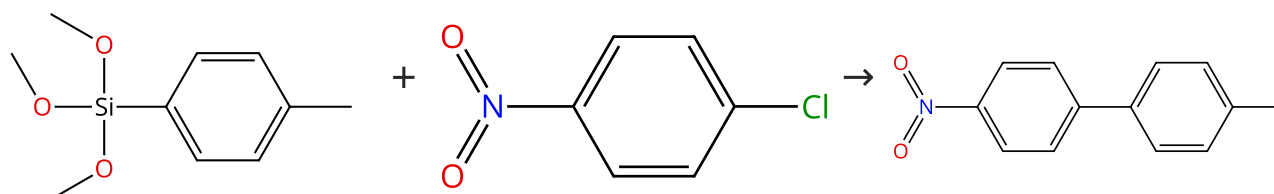
Suppliers (91)

Suppliers (68)

31-055-CAS-18274461	Steps: 1 Yield: 98%	Synthesis of adamantyl substituted 1,2,3-triazol-5-ylidene ligands and their PEPPSI-type palladium complexes
1.1 Reagents: Sodium hydroxide Catalysts: 1661862-34-5 Solvents: 1,4-Dioxane, Water; 4 h, 80 °C		By: Mitsui, Takashi; et al Tetrahedron (2015), 71(10), 1509-1514.
Experimental Protocols		

Scheme 25 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (51)

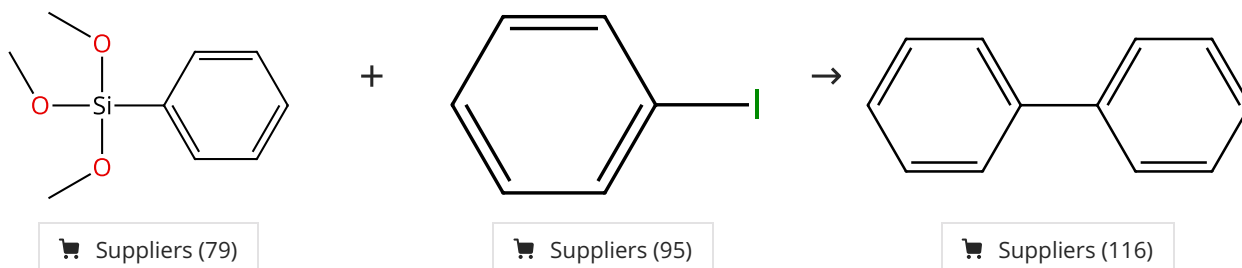
Suppliers (65)

Suppliers (42)

31-055-CAS-14454861 Steps: 1 Yield: 98% 1.1 Reagents: Tetrabutylammonium fluoride Catalysts: Tri-<i>o</i>-tolylphosphine, Palladium, bis(acetonitrile) dichloro-; 40 min, 80 °C Experimental Protocols	Improved procedure for palladium-catalyzed Hiyama cross-coupling reaction of aryl halides with aryltri methoxysilanes under solvent-free conditions By: Li, Jin-Heng; et al Synthesis (2006), (6), 969-974.
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Scheme 26 (2 Reactions)

Steps: 1 Yield: 98%



31-055-CAS-14374831 Steps: 1 Yield: 98% 1.1 Reagents: 1<i>H</i>-imidazolium, 3-butyl-1-methyl-, fluoride (1:1) Catalysts: Palladium diacetate, 1<i>H</i>-imidazolium, 3-(3-cyanopropyl)-1-methyl-, hexafluorophosphate(1-) (1:1) Solvents: Acetonitrile; 8 h, 70 °C Experimental Protocols	Phosphane-Free Hiyama Cross-Coupling of Aryl and Heteroaryl Halides Catalyzed by Palladium Nanoparticles in Ionic Liquids By: Premi, Chanchal; et al European Journal of Organic Chemistry (2013), 2013(24), 5493-5499.
31-055-CAS-5372110 Steps: 1 Yield: 98% 1.1 Reagents: Tetrabutylammonium fluoride Catalysts: Palladium acetylacetonate, PEP 36 Solvents: <i>p</i>-Xylene; rt → 80 °C; 12 h, 80 °C	A Colorimetric High-Throughput Screening Method for Palladium-Catalyzed Coupling Reactions of Aryl Iodides Using a Gold Nanoparticle-Based Iodide-Selective Probe By: Jung, Eunhye; et al Angewandte Chemie, International Edition (2011), 50(19), 4386-4389, S4386/1-S4386/16.